

Session-14: Collection Processing & Transformation Operations –

Largest Number:

Program:

```
class Solution {  
  
    public String largestNumber(int[] nums) {  
  
        String[] arr = new String[nums.length];  
  
        for(int i=0;i<nums.length;i++){  
  
            arr[i] = String.valueOf(nums[i]);  
  
        }  
  
        Arrays.sort(arr, (a,b) -> (b+a).compareTo(a+b));  
  
        if(arr[0].equals("0")) return "0";  
  
        StringBuilder sb = new StringBuilder();  
  
        for(String s : arr){  
  
            sb.append(s);  
  
        }  
  
        return sb.toString();  
  
    }  
}
```

Output:

The screenshot displays a coding platform interface for the '179. Largest Number' problem. The left panel shows the problem description, which asks to arrange a list of non-negative integers to form the largest possible number. It includes two examples: Example 1 with input [10, 2] and output '210', and Example 2 with input [3, 30, 34, 5, 9] and output '9534330'. Constraints specify that the array length is between 1 and 100, and each element is between 0 and 10^9. The right panel shows the code editor with the provided Java solution, which sorts the array using a custom comparator and concatenates the elements. Below the code, the 'Test Result' section shows 'Accepted' status with a runtime of 1 ms. A table of test cases follows, with 'Case 1' selected, showing an input of [10, 2] and an output of '210', matching the expected result.

179. Largest Number Solved ✓

Medium Topics Companies

Given a list of non-negative integers `nums`, arrange them such that they form the largest number and return it.

Since the result may be very large, so you need to return a string instead of an integer.

Example 1:
Input: `nums = [10, 2]`
Output: `"210"`

Example 2:
Input: `nums = [3, 30, 34, 5, 9]`
Output: `"9534330"`

Constraints:

- $1 \leq \text{nums.length} \leq 100$
- $0 \leq \text{nums}[i] \leq 10^9$

Code

```
1 class Solution {  
2     public String largestNumber(int[] nums) {  
3         String[] arr = new String[nums.length];  
4         for(int i=0;i<nums.length;i++){  
5             arr[i] = String.valueOf(nums[i]);  
6         }  
7         Arrays.sort(arr, (a,b) -> (b+a).compareTo(a+b));  
8         if(arr[0].equals("0")) return "0";  
9         StringBuilder sb = new StringBuilder();  
10        for(String s : arr){  
11            sb.append(s);  
12        }  
13        return sb.toString();  
14    }  
15 }
```

Saved

Testcase **Test Result**

Accepted Runtime: 1 ms

Case 1 Case 2

Input

nums =
[10, 2]

Output

"210"

Expected

"210"